

A New Variance Ratio Test of Random Walk in Emerging Markets: A Revisit

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Key Words: Emerging stock markets, Random walk hypothesis, Middle East and North Africa (MENA) stock markets.

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Abstract

Using Wright's (2000) new non-parametric variance-ratio (VR) test, we revisit the empirical validity of the random walk hypothesis in eight emerging markets in the Middle East and North Africa (MENA): Bahrain, Egypt, Jordan, Kuwait, Morocco, Oman, Saudi Arabia, and Tunisia. We show that: (i) while returns from raw data do not follow a random walk, one cannot reject the random walk hypothesis, particularly when the data have not been corrected to remove any measurement bias arising from thin and infrequent trading prevalent in nascent and small stock markets; (ii) Wright's non-parametric VR is more appropriate for emerging stock markets; and (iii) our findings can explain the contradictory results in the literature regarding the relative efficiency of emerging markets in MENA.

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1. Introduction

Central to investors and policymakers dealing with emerging equity markets is the knowledge of how efficiently those markets incorporate market information into securities prices. Specifically, what are the empirical validations of the random walk hypothesis (RWH) in these markets? This study examines the question for eight emerging markets in the Middle East and North Africa (MENA): Bahrain, Egypt, Jordan, Kuwait, Morocco, Oman, Saudi Arabia, and Tunisia.

Equity markets in MENA provide an excellent case study on the efficiency of emerging stock markets for five prime reasons. First, the eight emerging markets have experienced a rapid growth in their market capitalization in recent years: Over a 10-year period ending December 2003, their total market capitalization almost quadrupled from US\$72.5 billion to more than US\$288 billion. Second, the opening of many MENA emerging markets to foreign investors in the 1990s has provided good opportunities, particularly to foreign investors seeking risk diversification (Azzam, 2002). Third, the MENA markets are quite heterogeneous in their size and accessibility to foreign investors. By the end of 2003, the Saudi stock market was the largest based on its market capitalization of approximately \$157.3 billion, while the Tunisian market was the smallest at approximately \$2.4 billion.¹ Generally, the markets of Egypt, Jordan, Morocco, and Tunisia are open to foreign investors with few restrictions, but those of Bahrain,

¹ Measured by the number of listed companies, Egypt has the biggest stock market in the region with 967 companies listed while Bahrain is the smallest with 44 companies listed. Table 1 provides more information about each individual market.

Kuwait, Oman, and Saudi Arabia remain largely closed to most foreign investors.² As a result, there are few financial cross-links amongst the MENA stock markets even though they are geographically close together (see Girard and Ferreira, 2004). Fourth, the risk-return relationships of stocks listed on MENA emerging markets are quite unique, if not anomalous. They exhibit low returns and low volatility compared to high returns and high volatility generally observed in emerging markets in Asia, Latin America, and East Europe.³ Last, despite their recent rapid growth, several MENA markets, such as Morocco, Oman, and Tunisia, have been largely ignored in the literature concerned with the efficiency of emerging markets. Of the few published studies, which we are able to ascertain, the empirical findings on the random walk in MENA markets have thus far been inconclusive. It is the aim of this paper to make a complementary contribution to this important issue relating to those eight emerging markets in MENA which are of growing importance to investors and policy makers world-wide.

The principal tools used for the diagnostic testing of the RWH in emerging markets include the Lo-MacKinlay (1988) variance ratio (VR) test, the Chow-Denning (1993) VR test, the unit root, ARIMA, GARCH, artificial neural network tests, and the bootstrap test. Hoque, Kim, and Pyun (2006) report that, of 18 identified published studies on RWH in emerging stock markets, 16 use the Lo-MacKinlay or Chow-Denning VR tests along with other tests. Recently, Wright's (2000) rank and sign non-parametric VR test, initially developed for foreign exchange markets, is used in the study of RWH in stock prices (Bugak and Brorsen, 2003; Hoque, Kim, and Pyun, 2006; and Belaire-Franch and Opong, 2005b).

² For certain investments, investors from neighboring Arab countries are not considered as foreign investors (*The Arab Monetary Fund* (2003b, c)).

³ See Girard and Ferreira (2004). Other studies such as Bakaert and Harvey (1995) and Amoateng and Kargar (2004) suggest that MENA markets are heavily influenced by local investors who concentrate the investment decision on local economic and market conditions. See Smith and Ryoo (2003) for the RWH in emerging markets in East Europe.

In this study, we employ the Lo-MacKinlay VR, Wright's rank and sign VR, and the standard runs tests.⁴ We conclude that: (i) the raw return series in all eight MENA stock markets do not follow a random walk process when evaluated by Wright's (2000) rank and sign VR test; (ii) once the returns data are adjusted to reconcile distortions from thinly and infrequently traded stocks, random walk in all eight stock markets cannot be rejected by the rank and sign VR test; (iii) when these findings are compared to the results obtained by the Lo-MacKinlay VR and the standard runs tests, Wright's rank and sign test yields considerably stronger evidence of random walk in all eight markets.

2. Brief literature survey

Random walk properties of the time series of stock prices have long been prominent in the study of the stock return generating process.⁵ A number of studies have attempted to address the issue of random walk and the relative efficiency of MENA emerging markets but have achieved mixed results. Butler and Malaikah (1992) find evidence of inefficiency in the Saudi Arabian stock market, but not in the Kuwaiti market. Dickinson and Muragu (1994) find the Nairobi stock market to be efficient but El-Erian and Kumar (1995) find the Turkish and the Jordanian markets to be inefficient. Antoniou and Ergul (1997) find the Turkish stock market efficient at high trading volumes. Abraham, Seyyed, and Alsakran (2002) examine the random walk in three Gulf markets (Saudi Arabia, Kuwait, and Bahrain) and find that the stock markets

⁴ Due to space limitations, the Chow-Denning results are not included. The essential difference between the Lo-MacKinlay and Chow-Denning tests is that the former covers a single holding period whereas the latter simultaneously tests the equality of the VR to one over multiple holding periods (see Section 3). Hoque, Kim, and Pyun (2006) show that the two diagnostic tests yield similar results for RWH in Asian emerging stock markets when they are compared with testing results obtained with Wright's rank and sign VR test. Wright (2000) reports that the rank and sign VR test is more powerful than the Lo-MacKinlay and Chow-Denning tests of the RWH in foreign exchange markets.

⁵ For further discussions on random walk, see Summers (1986), Fama and French (1988), Lo and MacKinlay (1988), Liu and He (1991), and Malkiel (2005).

of Saudi Arabia and Bahrain, but not Kuwait, are efficient. The interdependence of the Istanbul Stock Exchange with the global markets is examined by Darrat and Benkato (2003). The weak-form efficiency of the Casablanca Exchange is studied by Ghysels and Cherkaoui (2003). Using Wright's (2000) non-parametric VR tests, Bugak and Brorsen (2003) find that the nonparametric tests provide evidence against the random walk in the Istanbul stock exchange.⁶

In non-MENA emerging markets, Barnes (1986) reports that the Kuala Lumpur stock market is inefficient. Panas (1990) adds that market efficiency cannot be rejected for the Greece market while Urrutia (1995) rejects the RWH for the markets of Argentina, Brazil, Chile, and Mexico markets. In contrast, Ojah and Karemera (1999) find that RWH holds in Argentina, Brazil, Chile, and Mexico. Grieb and Reyes (1999) re-examine the random walk properties of stocks traded in Brazil and Mexico using the VR test and conclude that the index returns in Mexico exhibit mean reversion and a tendency toward a random walk in Brazil. Alam, Hasan, and Kadapakkam (1999) examine the market efficiency for five Asian markets (Bangladesh, Hong Kong, Malaysia, Sri Lanka, and Taiwan) and conclude that all the index returns follow a random walk process with the exception of Sri Lanka. Darrat and Zhong (2000) and Poshakwale (2002) reject the RWH for the Chinese and Indian stock markets, respectively. Recently, Hoque, Kim, and Pyun (2006) test the RWH for eight emerging markets in Asia using Wright's (2000) rank and sign VR test and find that stock prices of most Asian developing countries do not follow a random walk with the possible exceptions of Taiwan and Korea.⁷

In this study, we show that Wright's non-parametric VR test is more powerful than either the Lo-MacKinlay parametric VR test or the conventional runs test over a wide range of time

⁶ While the Istanbul Stock Market may sometimes be regarded as one of the MENA markets, it is not included in our study.

⁷ With Wright's method, Belaire-Franch and Opong (2005b) find random walk behaviors in UK's FTSE index return series.

series. These series display diagnostic traits such as serial correlation, non-normal distribution, and conditional heteroskedasticity, all of which are known to lead to erroneous statistical inference *a priori* especially when the sampling distribution of statistics being tested is far from normal in the study of finite samples.⁸

3. Data and methodology

3.1. Data

This study covers eight equity markets in the Middle East and North Africa (MENA). In order to avoid any possible day-of-the-week effect, we use Wednesday data obtained from the Arab Monetary Fund (2003a). If an exchange is closed on Wednesday, Tuesday's data are used. Nominal stock returns weighted by market capitalization are calculated as the logarithmic difference in the stock prices in U.S. dollars between two consecutive periods. Weekly data for each individual market are obtained from Bahrain, Egypt, Jordan, Kuwait, Morocco, Oman, Saudi Arabia, and Tunisia from October 1994 through December 2003 except for Egypt where the data series start from January 1996.

Descriptive statistics on the weekly index returns of all eight MENA countries are reported in Table 1. The Kuwaiti market has the highest mean weekly return (0.19%), while the Tunisian market has the lowest (-0.27%). However, Egypt has the highest standard deviation (3.5%), and Bahrain the lowest (1.59%). The Sharpe ratio indicates that the relative risk-return trade-off is similar in all markets, except for Oman (1.1%) and Tunisia (-9.8%).

[Insert Table 1 about here]

⁸ A more detailed exposition on the analytical advantages of Wright's rank and sign VR tests over other tests is presented in Section 3.2.

The return series of the eight markets exhibit significant levels of skewness and kurtosis. The skewness of the return series is positive for Jordan, Oman, and Tunisia but negative for all the others. The positive skewness implies that the stock index returns are flatter to the right compared to the normal distribution. The kurtosis reported for each country indicates that the stock return distributions have sharp peaks compared to a normal distribution. One inference from the skewness and kurtosis findings is that the return series of all eight markets are not likely normal distributions. In fact, Jarque-Bera's statistics for testing normality confirm the significant non-normality of returns in the eight MENA markets.

3.2. Comparison of parametric and non-parametric variance ratio tests

In the study of RWH in emerging markets, VR tests have been by far the most widely used econometric tools since the pioneering work of Lo and MacKinlay (1988).⁹ Chow and Denning (1993) modify the Lo-MacKinlay parametric VR so that a set of multiple VRs over a number of holding periods can be jointly tested. It is critical to note that both the Lo-MacKinlay and Chow-Denning tests are asymptotic tests, and that their sampling distributions are approximated by their limiting distributions.¹⁰ In other words, the assumption underlying the VR tests is that stock returns are at least identically, if not normally, distributed and that the variance of the random walk increments in a finite sample is linear in the sampling interval.

Even though the Lo-MacKinlay VR test is quite powerful in testing for homoskedastic or heteroskedastic *iid* nulls (Smith and Ryoo, 2003), it is important to note that the sampling distribution of the VR statistic can be far from normal in finite samples, showing severe bias and

⁹ The gist of VR tests is that if a stock's return is purely random, the variance of k -period return is k times the variance of one-period return. Hence, the VR, defined as the ratio of $1/k$ times the variance of the k -period return to the variance of the one-period return, should be equal to one for all values of k .

¹⁰ For example, the limiting distribution of the Lo-MacKinlay test statistic is standard normal, and the critical values associated with the standard normal distribution are used for statistical inference for finite samples.

right skewness. These finite sample deficiencies may give rise to serious size distortions or low power, which can lead to misleading inferences. This is especially true when the sample size is not large enough to justify asymptotic approximations.

To overcome these problems, Wright (2000) introduces the non-parametric “rank and sign” VR test that does not resort to asymptotic arguments when its sampling distribution is approximated. They are thus more powerful tests than the conventional VR tests when return data are highly non-normal and non-stationary (Wright, 2000). These points are significant, as the sizes of MENA stock exchanges are relatively small, and the trading patterns of the exchanges are marked by infrequent and thin trading.¹¹

Since we are adopting individual VR tests in a joint hypothesis of random walk, Wright’s VR test is susceptible to size distortions. Following the modification employed by Belaire-Franch and Opong (2005a) on Wright’ test, we apply the Sidack adjusted p -values and a bootstrapping technique to overcome the size distortion.¹²

Wright’s VR test is susceptible to a further error: It may lead to erroneous inferences because the interaction of noise traders with other traders may generate non-linear deterministic systems at times which may give rise to random walk in stock prices (Poshakwale, 2002). It should be noted that Wright’s test tests for linear dependency only. Thus, the evaluation of RWH should be confirmed by the absence of both linear and non-linear dependence, since rejection of linear dependence does not imply independence but merely suggests a lack of autocorrelation (Granger and Anderson, 1978).¹³ To evaluate linear and non-linear dependencies of observed data in MENA markets, we apply the Brock, Dechert, Scheinkman, and Le Baron’s

¹¹ The derivation of Wright’s non-parametric rank and sign VR test is available upon request.

¹² We are indebted to an anonymous referee who suggested that we address the issue of size distortions in a joint hypothesis test.

¹³ We thank an anonymous referee for suggesting the BDSL linear and non-linear dependency test. Also, please see Belaire-Franch and Opong (2005b) who evaluated Wright’s VR test with the BDSL test.

(1996; BDSL hereafter) test on the residuals of the ARMA model as suggested by Hsieh (1991). In a nutshell, the null hypothesis under the BDSL test is that the data are identically and independently distributed.¹⁴ If the hypothesis is rejected, there is a high probability that the time series is nonlinear or has chaotic characteristics.

3.3. *Runs test*

Until Wright's (2000) work on the rank and sign VR test, the runs test was the most commonly used non-parametric test of the RWH. It does not require that return distributions are normally or identically distributed, the condition that most stock return statistics cannot satisfy. Moreover, it eliminates the effect of extreme values often found in the return data.¹⁵

In testing the efficiency of emerging markets, it is necessary to take into account thin trading that typically characterizes these markets. Thin (or infrequent) trading occurs when stocks do not trade at every consecutive interval, and thin or infrequent trading of stocks is known to give rise to statistical biases in the time series of stock prices.¹⁶ The bias of thinly traded shares is caused by prices that are recorded at the end of a time period that has a tendency to represent an outcome of a transaction occurring prior to the period in question. This means that thin trading induces the problem of serial correlation.

To correct for the effect of thin trading, the methodology of Miller, Muthuswamy, and Whaley (1994) is employed. Essentially, it removes the effect of thin trading by a moving average (MA) that reflects the number of non-trading days and calculates returns adjusted for the

¹⁴ The BDSL test is capable of locating many types of departures from independent and identical distribution, such as non-stationarity, non-linearity, and deterministic chaos. It uses the concept of correlation dimensions (CD) proposed by Grassberger and Procaccia (1983). The CD technique is designed to reveal evidence of a non-linear structure in the data. We use embedding dimensions of 2 to 10 and epsilons ranging from 1/2 to 2 times the standard deviation (see Brock, Dechert, Scheinkman, and Le Baron (1996) for details).

¹⁵ The description of the runs test is available upon request.

¹⁶ A number of studies have investigated the impact of thin trading and discussed its consequences (see Lo and MacKinlay, 1989; Stoll and Whaley, 1990; and Miller, Muthuswamy, and Whaley, 1994).

effect of non-trading days. However, given the difficulties in identifying the non-trading days, Miller, Muthuswamy, and Whaley (1994) have shown that that is equivalent to estimating an AR(1) model from which the non-trading adjustment can be obtained. They assume that non-trading adjustment is constant over time. While this assumption may be correct for highly liquid markets, it is not the case for emerging markets. In testing for the RWH, all of our tests (two VR tests and the runs test) are implemented by using corrected returns.

4. Empirical results

4.1. Variance Ratio Tests Based on Ranks and Signs

Under the null hypothesis that stock returns follow a random walk, the variance ratios are expected to equal one.¹⁷ The results of the VR tests based on ranks and signs for the observed and corrected market data are reported in Tables 2 and 3, respectively.

[Insert Table 2 about here]

Table 2 shows that the estimates of the VR and their associated tests statistics, $R_1(k)$, $R_2(k)$, and $S_1(k)$ for lags $k = 2, 4, 8, 16$, and 32 are different from one at the 5% level of significance for all markets studied. When the raw observed data are used, the RWH is rejected for all MENA markets under the assumptions of both homoskedasticity and heteroskedasticity. As shown below, this rejection of the hypothesis is erroneous and misleading to investors and policy makers in these markets.

We repeat the analysis by correcting the returns for thin trading. The rank and sign variance ratios, $VR(k)$, the homoskedastic test statistics, $R_1(k)$, the heteroskedasticity-consistent test statistics $R_2(k)$ and $S_1(k)$ for the corrected returns in MENA markets, are reported in Table 3.

¹⁷ The VR is computed for multiples of 2, 4, 8, 16, and 32 weeks with one week return used as the base. The $VR(k)$, the homoskedastic test statistics, $Z_1(k)$, and the heteroskedastic-consistent test statistics $Z_2(k)$ for the MENA markets are not reported here but are available upon request.

The results indicate that the estimated rank and sign VR tests are not significantly different from one at all lags, suggesting that the RWH cannot be rejected for these markets. Even when heteroskedasticity is considered, the conclusions remain the same with not a single rejection of the null hypothesis of random walk.¹⁸ Once again, when the raw observed data are used, the RWH is rejected. But, when corrected for thin trading, the hypothesis cannot be rejected for all markets.

[Insert Table 3 about here]

In summary, the stock markets in the MENA region do not follow a random walk unless the published return data are corrected for thin trading. Our findings are quite crucial in showing that thin trading and non-synchronous prices do significantly affect the conclusions that can be drawn from statistical testing of the RWH in many emerging markets.

Chow and Denning (1993) point out that VR tests at differing intervals are susceptible to the effect of size distortion, which in turn leads to rejection rates larger than the nominal size. Belaire-Franch and Opong (2005a) show that rejection may increase for Wright's rank tests in particular when their test statistics are not robust in the presence of heteroskedasticity. To minimize the effect of size distortion, we apply Belaire-Franch and Opong's three alternative models. Specifically, we compute Wright's rank and sign VR test for two k values (2 and 4) when $N = 100$, three k values (2, 4, and 8) when $N = 200$ and four k values (2, 4, 8, and 16) when $N = 400$, and analyze the impact on the empirical size of each individual test.¹⁹ Further, we employ Monte Carlo simulation to produce the rates of rejection for Wright's VR test at the 5%

¹⁸The results of S_2 are available from the authors upon request.

¹⁹ We do not use the rolling window method used by Belaire-Franch and Opong's (2005b) as our weekly data for individual markets covering the 10-year period are insufficient for the rolling window methods.

significance level. In addition, in order to mitigate the effect of possible correlations that may exist among the statistics, we compute the bootstrap-adjusted p -values.²⁰

Using unadjusted raw data, Table 4 and 5 display the Sidack and bootstrapping-adjusted p -values for Wright's VR tests (R_1), (R_2), and (S_1), respectively. The results in Table 4 and 5 indicate that the null hypothesis of random walk in all eight MENA markets is rejected. However, when the data are corrected for thin trading, as Table 6 and 7 show, the null hypothesis cannot be rejected for all the markets. Note that our results do not vary significantly when we adjust for the effects of size distortion.

[Insert Tables 4-7 about here]

As discussed in section 3.2 above, the evaluation of RWH with Wright's VR test should be confirmed by the absence of both linear and non-linear dependence of the observed time series. Table 8 shows that the BDSL statistics for non-linear dependence are not significant at all levels for the eight MENA markets. This means that the null hypothesis of *iid* is not rejected in all the markets. Failure to reject *iid* confirms the absence of non-linear dependence in weekly returns.²¹ The absence of the non-linear dependence means the RWH cannot be rejected for the MENA markets.

[Insert Table 8 about here]

Our findings relating to the markets in Bahrain and Saudi Arabia are generally consistent with Abraham, Seyyed, and Alskran (2002), who confirm that these two MENA markets are weak-form efficient. However, our findings for the markets in Kuwait, Jordan, Egypt, and Morocco are inconsistent with those reported in Abraham, Seyyed, and Alskran (2002), El-Erian

²⁰ Due to space limitation, the tables of rejection probabilities and bootstrap-adjusted p -values of the three models are not reported, but are available from the authors upon request.

²¹ Poshakwale and Murinde (2001) applied the BDSL test to investigate the behavior of Hungarian and Polish markets.

and Kumar (1995), and Smith, Jefferies, and Ryoo (2000); these three studies reject the weak-form efficiency in the four MENA markets.

On balance, our findings contradict all previous results on the weak form efficiency of MENA markets except for two countries – Bahrain and Saudi Arabia. Even for these two stock markets, only the study by Abraham, Seyyed and Alskran (2002) affirms our findings when they corrected their data for the effect of thin trading for these two markets. Given the improved size and power properties of Wright's (2000) rank and sign test, our findings strongly suggest that the empirical validity of the weak-form efficiencies of MENA markets already reported in the literature should be re-examined, as many of these studies may have drawn erroneous statistical inferences in their studies based on the Lo-MacKinlay or Chow-Denning tests.

4.2. Runs Test

Results of the runs test are shown in Table 9, both for the observed indices and for the indices corrected for infrequent trading. In Panel A of the table for the observed raw indices, the actual number of runs (R) in each of the MENA markets is less than the expected number of runs under the null hypothesis of stock return independence. The resulting negative Z values for the MENA markets indicate positive serial correlation. The runs test shows that the successive returns for all countries under study are not independent at the 5% significance level. In Panel B, we show that, when the indices are corrected for infrequent trading, the results are strikingly different. In all eight countries, the expected and the actual number of runs are so close as to be virtually indistinguishable. Based on the corrected indices, one cannot reject the weak form market efficiency in any of the eight MENA equity markets. The results show that correcting for

thin trading with non-synchronous prices leads to an absolute reversal in the inferences on market efficiency.

[Insert Table 9 about here]

5. Summary and concluding remarks

With the new rank and sign test, we have revisited the empirical validity of the weak-form efficient market hypothesis for the stock markets in eight MENA countries: Bahrain, Egypt, Jordan, Kuwait, Morocco, Oman, Saudi Arabia, and Tunisia. Employing the new tests, our findings are compared to those obtained from the Lo-MacKinlay (1988) VR test and the runs test – two most widely used methods for testing the existence of random walk in stock returns.

Our findings show that the return behaviors of all eight MENA markets computed from the published index return series in their raw data form do not follow a random walk pattern. This may lead one to conclude that weak-form efficiency does not exist in these markets. However, a remarkable phenomenon uncovered in our study is that, when returns from the published indices are corrected for the statistical biases residing within the published indices, we cannot reject the RWH for any of the eight markets. All three testing methods implemented – rank and sign test, runs test, and the conventional VR test – have given rise to the same inference that all eight individual equity markets are weak-form efficient in processing price information relevant to their individual respective markets. Ancillary to this conclusion is the importance of recognizing the statistical quirk inherent in the published indices in the eight markets that arise structurally from thin and infrequent trading and which produces high serial correlations in the unadjusted published indices.

Our findings help explain the contradictory results in the literature regarding the relative efficiency of stock markets in the region. For instance, Abraham, Seyyed, and Alsakran (2002) found that the stock markets in Saudi Arabia and Bahrain were efficient, but the Kuwaiti market was not. Our findings are that, while the Saudi and Bahraini markets are much more robust statistically, the Kuwaiti market is weak-form efficient within the bounds of the critical statistics by the rank and sign test. It should be emphasized that the rank and sign VR test is an improved non-parametric test that is more powerful when compared to other non-parametric runs test used by Abraham, Seyyed, and Alsakran. More importantly, the conventional VR tests used by many researchers are inappropriate because returns from the indices are neither normally nor identically distributed, which is the crucial assumption upon which conventional VR tests are based.

Our findings have practical implications for policy makers in the region, in general, and for both individual and institutional investors, in particular. First, recent financial liberalization programs implemented by the MENA countries have been successful in making their domestic economies grow faster along with more vibrant stock markets. It is in the interest of the MENA nations to accelerate their financial reform measures, including privatization programs, so as to make their stock markets more transparent and efficient in processing information. Second, despite the current political uncertainty surrounding the war in Iraq, the fundamentals of the MENA equity markets are positive and the markets offer attractive investment opportunities for both domestic and overseas investors with reasonably good risk-adjusted returns.

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Table 1

Summary Descriptive Statistics of Weekly Returns for MENA Stock Markets

This table describes the summary statistics for the eight stock markets of Bahrain, Egypt, Jordan, Kuwait, Morocco, Oman, Saudi Arabia, and Tunisia.

	Bahrain	Egypt	Jordan	Kuwait	Morocco	Oman	Saudi Arabia	Tunisia
Exchange Traded	Bahrain Stock Exchange	Egypt Herms Index	Amman Financial Market	Kuwait General Index	Morocco Stock Exchange	Muscat Stock Exchange	Saudi General Index	Tunisia General Index
Time Period Covered	Oct. 1994-Dec. 2003	Jan. 1996-Dec. 2003	Oct. 1994-Dec. 2003	Oct. 1994-Dec. 2003	Oct. 1994-Dec. 2003	Oct. 1994-Dec. 2003	Oct. 1994-Dec. 2003	Oct. 1994-Dec. 2003
No. of listed companies	44	967	161	108	52	141	70	45
Market capitalization (Millions \$)	9,701.77	27,847.48	10,962.98	59,528	13,050.18	7,246.23	157,306.44	2,439.55
Volume trade (Millions \$)	261.14	4,349.12	2,607.14	54,728.88	2,443.46	1,334.3	159,055.55	188.53
Turnover ratio (%)	2.69	15.62	23.78	24.83	91.93	18.41	101.11	2.16
Mean returns	0.052	-0.20	0.11	0.19	0.15	0.03	0.13	-0.27
Max (R)	0.095	0.189	0.077	0.25	0.078	0.136	0.08	0.124
Min (R)	-0.125	-0.28	-0.066	-0.246	-0.087	-0.160	-0.096	-0.126
Std. Dev.	1.59	3.5	1.74	2.8	2.05	2.7	2.13	2.75
Sharpe Ratio	0.033	-0.057	0.063	0.068	0.073	0.011	0.061	-0.098
Skewness	-0.71 (0.00)	-1.11 (0.00)	0.43 (0.00)	-0.38 (0.00)	-0.06 (0.00)	0.29 (0.00)	-0.085 (0.00)	0.37 (0.00)
Kurtosis	16.3 (0.00)	17.0 (0.00)	5.07 (0.00)	30.2 (0.00)	5.5 (0.00)	9.5 (0.00)	5.2 (0.00)	8.0 (0.00)
Jarque-Bera	3,586 (0.00)	3,497 (0.00)	100.8 (0.00)	14,882 (0.00)	125 (0.00)	855 (0.00)	98 (0.00)	504 (0.00)
Observations	481	416	481	481	481	481	481	481

Notes:

(1) Weekly returns are computed as $R_t = \ln(P_t/P_{t-1})$

(2) Std. Dev. denotes standard deviation.

(2) Sharpe ratio is calculated as mean return over standard deviation.

(3) Numbers in parentheses below values for skewness, kurtosis, and Jarque-Bera report associated p -values for their significance.

(4) Turnover ratio is calculated by dividing the volume traded by market capitalization for the year.

Source: Arab Monetary Fund (2003a).

Table 2

Wright's Variance-Ratio Tests Results with Unadjusted Raw Data

This table reports the values of the Wright (2000) non-parametric variance ratio based on ranks and signs for the observed raw market data over k number of lags. The test statistics for the ranked-based tests (R_1 and R_2) and signed-based test (S_1) are provided in parentheses.

	Bahrain	Egypt	Jordan	Kuwait	Morocco	Oman	Saudi Arabia	Tunisia
R_1								
$k=2$	1.41 (3.97)**	1.34 (4.33)**	1.36 (3.34)**	1.23 (2.91)**	1.35 (3.32)**	1.25 (3.69)**	1.24 (4.59)**	1.29 (4.29)**
$k=4$	1.67 (6.32)**	1.48 (4.76)**	1.52 (4.35)**	1.29 (3.14)**	1.35 (3.66)**	1.29 (3.92)**	1.39 (5.29)**	1.31 (4.19)**
$k=8$	1.92 (6.55)**	1.69 (5.24)**	1.66 (4.77)**	1.67 (3.88)**	1.49 (3.77)**	1.39 (4.22)**	1.61 (5.92)**	1.46 (4.59)**
$k=16$	2.22 (7.45)**	1.91 (5.66)**	1.87 (4.76)**	1.75 (3.94)**	1.72 (3.94)**	1.49 (4.66)**	1.76 (6.18)**	1.58 (4.81)**
$k=32$	2.55 (9.76)**	2.23 (6.29)**	1.93 (5.25)**	1.99 (4.21)**	1.89 (4.35)**	1.83 (4.92)**	1.88 (6.59)**	1.79 (5.18)**
R_2								
$k=2$	1.21 (3.19)**	1.24 (3.25)**	1.31 (3.09)**	1.13 (2.89)**	1.25 (2.98)**	1.19 (3.04)**	1.23 (3.26)**	1.25 (3.25)**
$k=4$	1.61 (5.38)**	1.45 (3.26)**	1.42 (3.35)**	1.26 (2.99)**	1.31 (3.16)**	1.24 (3.14)**	1.37 (4.36)**	1.29 (3.65)**
$k=8$	1.87 (5.55)**	1.65 (4.52)**	1.56 (3.37)**	1.57 (3.18)**	1.43 (3.27)**	1.37 (3.88)**	1.58 (4.39)**	1.45 (3.76)**
$k=16$	2.02 (6.55)**	1.89 (4.76)**	1.67 (3.76)**	1.71 (3.69)**	1.67 (3.69)**	1.43 (4.16)**	1.73 (5.42)**	1.56 (3.98)**
$k=32$	2.45 (7.46)**	2.23 (4.86)**	1.89 (4.35)**	1.98 (3.97)**	1.86 (4.15)**	1.78 (4.21)**	1.83 (5.66)**	1.78 (4.56)**
S_1								
$k=2$	1.11 (3.87)**	1.14 (4.23)**	1.21 (3.64)**	1.09 (2.99)**	1.15 (3.32)**	1.09 (3.87)**	1.13 (4.66)**	1.15 (4.45)**
$k=4$	1.51 (6.22)**	1.35 (4.56)**	1.22 (4.75)**	1.16 (3.54)**	1.21 (3.86)**	1.14 (3.95)**	1.17 (5.46)**	1.32 (4.55)**
$k=8$	1.77 (6.45)**	1.45 (5.14)**	1.36 (4.97)**	1.17 (3.98)**	1.43 (3.97)**	1.27 (4.82)**	1.38 (5.94)**	1.25 (4.86)**
$k=16$	1.67 (7.15)**	1.59 (5.86)**	1.37 (4.86)**	1.21 (3.99)**	1.27 (4.09)**	1.13 (4.96)**	1.53 (6.42)**	1.29 (4.88)**
$k=32$	2.15 (9.16)**	1.53 (6.86)**	1.29 (5.45)**	1.58 (4.31)**	1.36 (4.65)**	1.38 (5.09)**	1.63 (6.96)**	1.58 (5.76)**

** Significant at the 5% level.

Table 3

Wright's Variance-Ratio Tests Results with Corrected Data

This table reports the values of the Wright (2000) non-parametric variance ratio based on ranks and signs over k number of lags for data over that has been corrected for thin trading. The test statistics for the ranked-based tests (R_1 and R_2) and signed-based test (S_1) are provided in parentheses.

	Bahrain	Egypt	Jordan	Kuwait	Morocco	Oman	Saudi Arabia	Tunisia
R_1								
$k=2$	1.02 (0.76)	0.99 (0.97)	1.05 (1.15)	0.95 (1.36)	1.03 (0.75)	1.06 (1.35)	0.99 (1.23)	0.96 (0.75)
$k=4$	1.01 (0.88)	0.95 (1.23)	1.06 (1.26)	0.96 (1.17)	0.96 (0.98)	1.07 (1.18)	1.06 (1.19)	1.05 (0.98)
$k=8$	1.02 (1.01)	1.04 (1.15)	1.08 (1.24)	1.04 (1.31)	1.07 (1.19)	1.08 (1.46)	1.08 (1.24)	1.06 (1.17)
$k=16$	1.08 (1.08)	1.06 (1.25)	1.05 (1.26)	1.06 (1.28)	1.05 (1.33)	1.07 (1.12)	1.11 (1.44)	1.08 (1.12)
$k=32$	1.06 (1.19)	1.01 (1.34)	1.06 (1.26)	1.08 (1.38)	1.07 (1.38)	1.09 (1.36)	1.10 (1.16)	1.09 (1.33)
R_2								
$k=2$	0.99 (0.71)	0.97 (0.90)	1.01 (1.11)	0.98 (1.26)	0.99 (0.55)	1.02 (1.26)	0.98 (1.17)	0.99 (0.65)
$k=4$	1.04 (0.80)	0.99 (1.19)	1.01 (1.16)	1.05 (1.12)	0.98 (0.88)	1.04 (1.11)	1.01 (1.09)	1.01 (0.88)
$k=8$	1.04 (0.99)	1.01 (1.10)	1.03 (1.14)	1.02 (1.25)	1.04 (1.09)	1.01 (1.36)	1.04 (1.18)	1.02 (1.07)
$k=16$	1.04 (1.01)	1.03 (1.15)	1.02 (1.17)	1.05 (1.21)	1.02 (1.23)	1.03 (1.08)	1.10 (1.32)	1.03 (1.08)
$k=32$	1.05 (1.11)	1.04 (1.24)	1.03 (1.15)	1.07 (1.28)	1.03 (1.27)	1.07 (1.29)	1.11 (1.03)	1.06 (1.23)
S_1								
$k=2$	0.99 (0.86)	1.02 (0.97)	1.03 (1.25)	0.98 (1.33)	0.99 (0.85)	0.98 (0.95)	0.96 (1.16)	1.02 (1.25)
$k=4$	1.03 (1.18)	0.98 (1.17)	1.01 (1.27)	1.07 (1.38)	0.98 (0.98)	1.05 (1.08)	1.07 (1.29)	1.06 (1.28)
$k=8$	1.02 (1.32)	1.01 (1.15)	1.02 (1.42)	1.02 (1.41)	1.02 (1.19)	1.01 (1.16)	1.08 (1.34)	1.07 (1.27)
$k=16$	1.02 (1.28)	1.03 (1.31)	1.02 (1.42)	1.01 (1.31)	1.01 (1.23)	1.02 (1.36)	1.11 (1.44)	1.11 (1.32)
$k=32$	1.03 (1.32)	1.05 (1.29)	1.03 (1.42)	1.10 (1.29)	1.04 (1.28)	1.03 (1.26)	1.26 (1.36)	1.14 (1.43)

Table 4

Wright's VR tests Sidack-Corrected p -values with Unadjusted Raw Data

This table shows the Sidack-adjusted p -values of Wright's VR tests corresponding to a k lagged value of 4. The data used are the observed, unadjusted ones. R_1 and R_2 refer to the ranked-based tests and S_1 refers to the signed-based test.

	Bahrain	Egypt	Jordan	Kuwait	Morocco	Oman	Saudi Arabia	Tunisia
R_1	0.089*	0.045**	0.092*	0.063*	0.096*	0.055*	0.067*	0.065*
R_2	0.078*	0.076*	0.065*	0.086*	0.085*	0.092*	0.093*	0.074*
S_1	0.093*	0.097*	0.083*	0.041**	0.046**	0.063*	0.072*	0.082*

** Significant at the 5% level.

* Significant at the 10% level.

Table 5

Wright's VR Tests Bootstrap-Corrected p -values with Unadjusted Raw Data

This table shows the bootstrap-adjusted p -values of Wright's VR tests corresponding to a k lagged value of 4 with 1000 replications. The data used are the observed, unadjusted ones. R_1 and R_2 refer to the ranked-based tests and S_1 refers to the signed-based test.

	Bahrain	Egypt	Jordan	Kuwait	Morocco	Oman	Saudi Arabia	Tunisia
R_1	0.065*	0.062*	0.072*	0.093*	0.098*	0.091*	0.078*	0.092*
R_2	0.078*	0.089*	0.068*	0.034**	0.058*	0.034**	0.081*	0.071*
S_1	0.098*	0.072*	0.091*	0.052*	0.084*	0.025**	0.092*	0.083*

** Significant at the 5% level.

* Significant at the 10% level.

Table 6

Wright's VR tests Sidack-Corrected p -values with Corrected Data

This table shows the Sidack-adjusted p -values of Wright's VR tests corresponding to a k lagged value of 4. The data have been corrected for thin trading using the method of Miller, Muthuswamy, and Whaley (1994). R_1 and R_2 refer to the ranked-based tests and S_1 refers to the signed-based test.

	Bahrain	Egypt	Jordan	Kuwait	Morocco	Oman	Saudi Arabia	Tunisia
R_1	0.261	0.312	0.192	0.226	0.332	0.535	0.367	0.452
R_2	0.378	0.433	0.265	0.286	0.385	0.192	0.193	0.244
S_1	0.193	0.155	0.383	0.513	0.164	0.363	0.372	0.216

Table 7

Wright's VR Tests Bootstrap-Corrected p -values with Corrected Data

This table shows the bootstrap-adjusted p -values of Wright's VR tests corresponding to a k lagged value of 4. The data have been corrected for thin trading using the method of Miller, Muthuswamy, and Whaley (1994). R_1 and R_2 refer to the ranked-based tests and S_1 refers to the signed-based test.

	Bahrain	Egypt	Jordan	Kuwait	Morocco	Oman	Saudi Arabia	Tunisia
R_1	0.265	0.162	0.272	0.493	0.114	0.391	0.478	0.392
R_2	0.178	0.289	0.368	0.234	0.152	0.134	0.281	0.171
S_1	0.398	0.172	0.191	0.452	0.124	0.172	0.192	0.523

Table 8

BDSL Test for Corrected Weekly Return Series

This table shows the results of the BDSL test of Brock, Dechert, Scheinkman, and Le Baron (1996). It is used to evaluate linear and non-linear dependencies of the observed data. The null hypothesis under the BDSL test is that the data are identically and independently distributed. If the hypothesis is rejected, there is a high probability that the time series is nonlinear or has chaotic characteristics.

<i>m</i>	Bahrain	Egypt	Jordan	Kuwait	Morocco	Oman	Saudi Arabia	Tunisia
2	0.34	0.22	0.35	0.24	0.19	0.18	0.28	0.15
3	0.32	0.27	0.39	0.36	0.28	0.21	0.24	0.18
4	0.37	0.37	0.49	0.27	0.32	0.58	0.36	0.22
5	0.42	0.56	0.67	0.45	0.36	0.73	0.27	0.25
6	0.65	0.67	0.66	0.57	0.56	0.85	0.56	0.38
7	0.56	0.74	0.84	0.78	0.69	0.89	0.69	0.44
8	0.57	0.63	0.94	0.87	0.75	0.93	0.78	0.65
9	0.78	0.89	1.11	0.76	0.87	1.05	0.97	0.56
10	0.82	0.93	1.24	0.79	0.93	1.08	1.23	0.63

m: denotes dimensions

Critical values are 1.645 and 1.96 at the 10% and 5% significance levels, respectively.

Table 9

Runs Test Results with Market Index

This table provides the results of the runs test for both the observed indices (Panel A) and for the indices corrected for thin trading (Panel B). We report the actual number of runs (R), expected number of runs (m), and the number of positive, negative, and no-change runs in each of the MENA markets.

	Bahrain	Egypt	Jordan	Kuwait	Morocco	Oman	Saudi Arabia	Tunisia
<i>Panel A: Observed Raw Data</i>								
Observations (N)	481	416	481	481	481	481	481	481
n (+)	244	206	243	244	247	245	245	243
n (-)	237	210	238	237	234	236	236	238
n (0)	0	0	0	0	0	0	0	0
Expected runs (m)	250	210	246	245	249	251	242	244
Actual runs (R)	221	197	221	226	220	223	221	219
Standard error	2.45	3.39	3.87	2.74	3.05	3.35	2.78	4.24
Z-Statistics	-3.148*	-3.56*	-3.27*	-3.78*	-3.26*	-3.72*	-3.27*	-2.98*
<i>Panel B: Corrected Data</i>								
Observations (N)	480	415	480	480	480	480	480	480
n (+)	241	202	244	242	244	245	241	246
n (-)	239	213	236	238	236	235	239	234
n (0)	0	0	0	0	0	0	0	0
Expected runs (m)	239	199	237	236	241	238	235	237
Actual runs (R)	237	201	235	235	239	236	233	235
Standard error	2.41	3.59	3.27	2.63	2.85	3.28	2.18	4.76
Z-Statistics	-0.654	-0.457	-0.875	-0.845	-0.964	-0.637	-0.875	-0.745

Notes:

n(+), n(-), and (0) represent the number of successive sequence of positive, negative, and zero price change, respectively.

* Significant at the 5% level.